DIGITAL SYSTEMS

SCSC1013 - COMPUTER SYSTEM & ORGANIZATION

Module 1 · Logic, Data Representation & Binary Arithmetic

Dr Zuriahati Mohd Yunos · BSE (Hons.)

MODULE 1 · LEARNING ARC

Three weeks, one digital-foundations story

01	**SIGNALS**	Binary states and waveforms establish discrete information.
02	**LOGIC**	Gates and truth tables define two-state rules.
03	**REPRESENT**	Radix, width and codes assign meaning to bit patterns.
04	**CALCULATE & INTERPRET**	Fixed-width arithmetic separates results, carry, borrow and overflow.

The bits stay the same; the interpretation contract changes.

WEEK 1 OUTCOMES

By the end of Week 1, you can…

01	**Distinguish**	analog and digital representations
02	**Interpret**	binary states, levels and transitions
03	**Apply**	AND, OR, NOT, NAND, NOR, XOR and XNOR
04	**Translate**	symbol ↔ expression ↔ truth table ↔ trace

Four 50-minute blocks · practice is built into every block

Which examples behave like two-state signals?

EXAMPLE A

LIGHT SWITCH

OFF / ON

EXAMPLE B

ROOM TEMPERATURE

changes smoothly

EXAMPLE C

DOOR SENSOR

CLOSED / OPEN

EXAMPLE D

BATTERY LEVEL

changes over a range

Vote first. Then justify the representation you assumed.

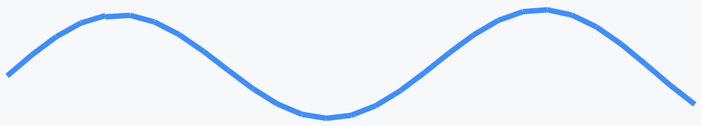
Analog varies; digital uses distinct states

RANGE

ANALOG

A value can move smoothly through a range.

Examples: temperature, light intensity, microphone voltage.

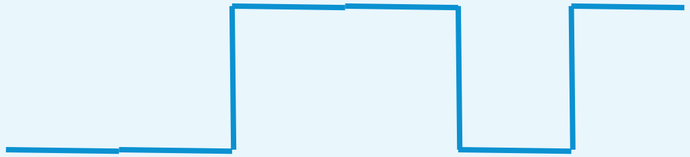


STATES

DIGITAL

A value belongs to defined discrete levels.

This course begins with two states: 0 and 1.



Representation is the key idea—not transistor or voltage-threshold theory.

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Names give binary states a precise meaning

0

LOW

FALSE · inactive · OFF

1

HIGH

TRUE · active · ON

A, B

named inputs

DIGITAL RULE

defines the relationship

Y

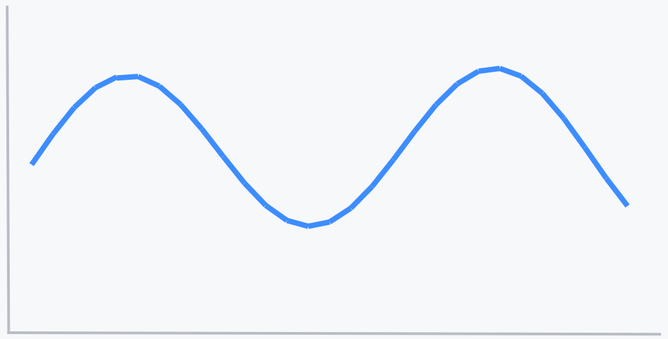
named output

Always define the application meaning: for example, D = 1 means ‘door open’.

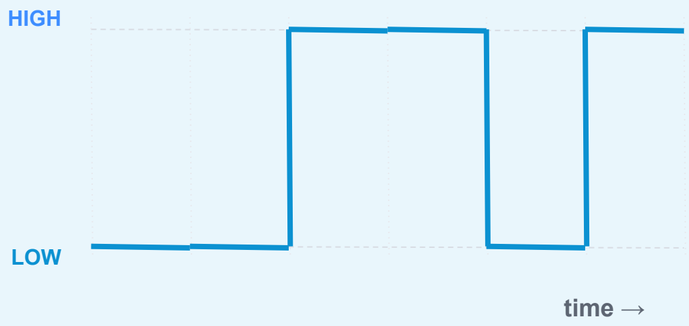
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A digital waveform keeps only defined levels

CONTINUOUS SIGNAL



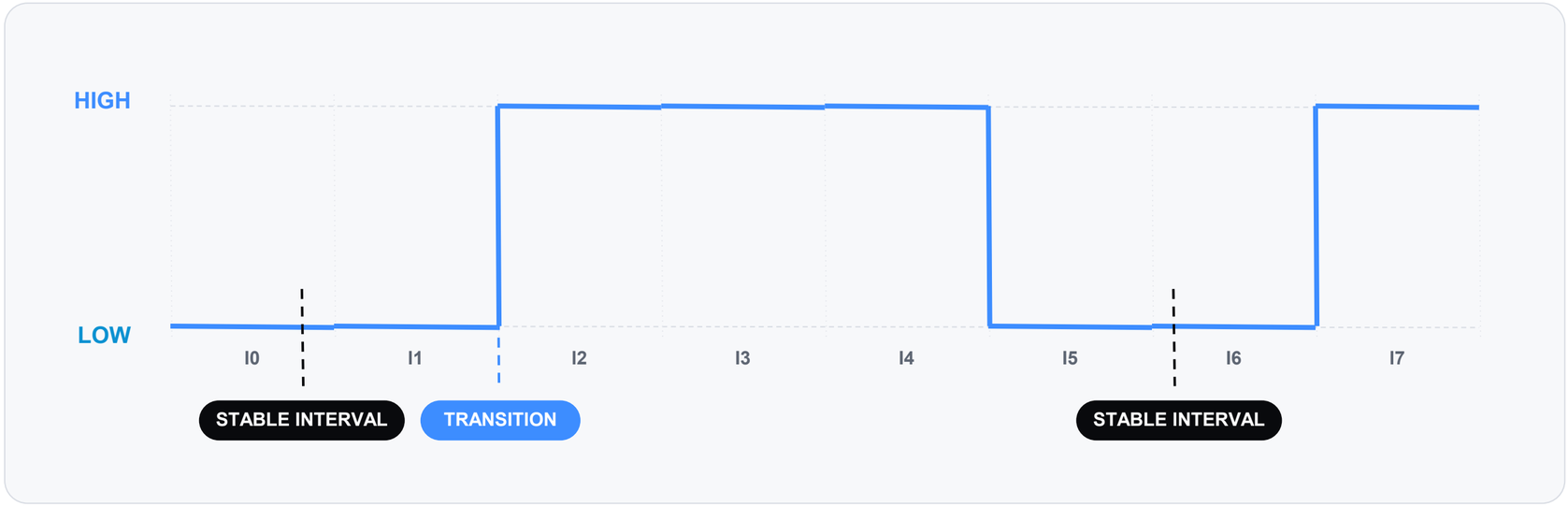
TWO-LEVEL REPRESENTATION



Two levels simplify each moment to a clear logic state: 0 or 1.

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Read a waveform from left to right



At any interval, read the level first. Name edge directions later in Block 4.

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Worked example: door sensor → warning lamp



D	L	MEANING
0	1	door closed → lamp off
1	0	door open → lamp on

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PAIR ACTIVITY · 12 MIN

Design one simple two-state system

1

Choose

an everyday system

2

Name

one input and one output

3

Define

0 and 1 for both signals

4

Sketch

input → rule → output

CHECK **Two states are genuine · labels are unambiguous · arrows show direction**

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A clear digital-system sketch removes ambiguity



P: 0 absent · 1 present

L: 0 off · 1 on

MODEL CHECK

- ✓ Signals have names
- ✓ Every 0/1 meaning is defined
- ✓ Input and output are distinct
- ✓ Rule is visible
- ✓ Arrows show data flow

Next: gates define the controller's rule.

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When should the warning lamp turn on?

INPUT 1

E = SYSTEM ENABLED

0: disabled
1: enabled

INPUT 2

D = DOOR OPEN

0: closed
1: open

The lamp turns on only when

E = 1 AND D = 1

E	D	L
0	0	?
0	1	?
1	0	?
1	1	?

Predict all four rows before naming the gate.

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AND requires every condition to be true

FUNCTION

Output 1 only when both inputs are 1.

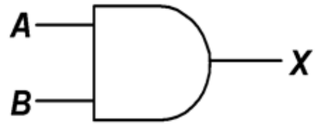


IEC functional block symbol

BOOLEAN

$$Y = A \cdot B$$

SYMBOL



TRUTH TABLE

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

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OR requires at least one true condition

FUNCTION

Output 1 when A, B, or both are 1.

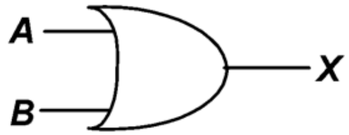


IEC functional block symbol

BOOLEAN

$$Y = A + B$$

SYMBOL



TRUTH TABLE

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

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NOT reverses one logic state

FUNCTION

The output is the opposite of the input.



IEC functional block symbol

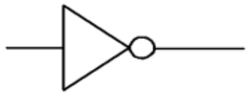
BOOLEAN

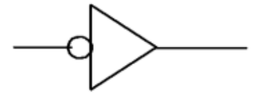
$Y = A'$

TRUTH TABLE

A	Y
0	1
1	0

SYMBOL





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A truth table tests every input combination

For two inputs, use the same row order every time:

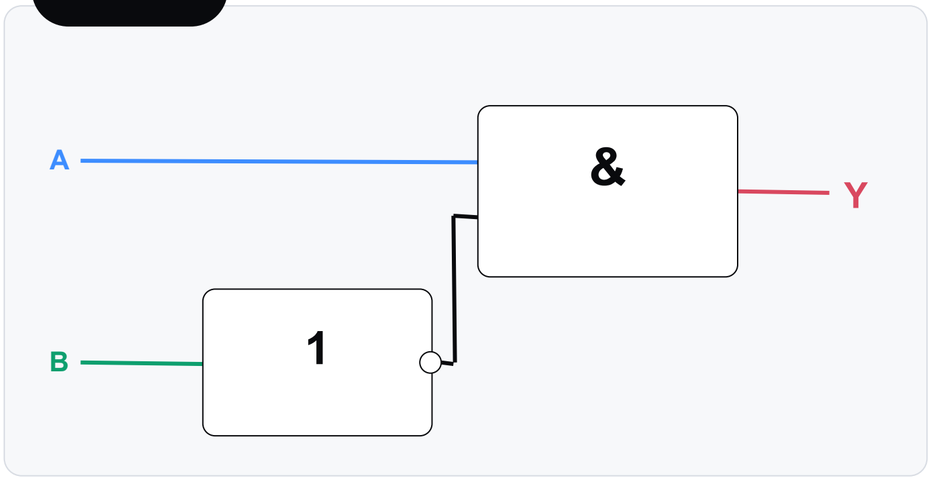
00	01	10	11
A=0 B=0	A=0 B=1	A=1 B=0	A=1 B=1

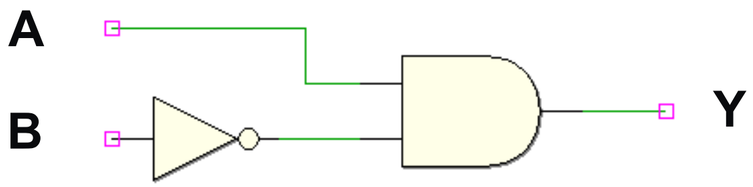
A	B	AND	OR
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

Apply the verbal rule to each row—do not guess from the gate name.

Symbols, expressions, and tables must agree

CIRCUIT





EXPRESSION

$$Y = A \cdot B'$$

Evaluate the intermediate value first:

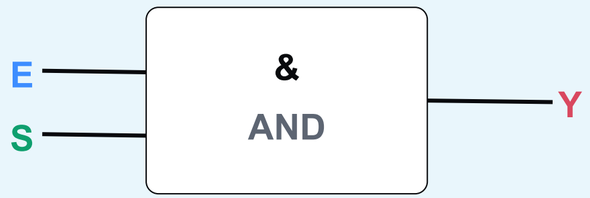
1. Find B’
2. Combine A AND B’
3. Record Y

A	B	B’	Y
0	0	1	0
0	1	0	0
1	0	1	1
1	1	0	0

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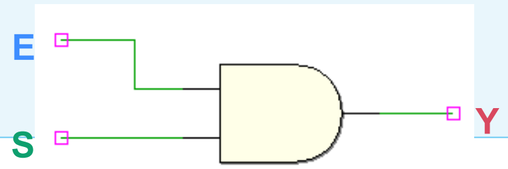
Worked example: enable AND sensor

$$Y = E \cdot S$$



E = 1: system enabled

S = 1: sensor active



E	S	Y	PHYSICAL MEANING
0	0	0	disabled; idle
0	1	0	disabled; sensor active
1	0	0	enabled; idle
1	1	1	enabled; sensor active

Only 11 produces Y = 1.

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INDIVIDUAL → PAIR → CLASS · 12 MIN

Match, calculate, and justify

A **MATCH**

Each basic gate ↔ its expression ↔ its truth table.

B **CALCULATE**

For Y = A + B’, find all four outputs in row order 00, 01, 10, 11.

C **JUSTIFY**

Choose one row and explain the output in a complete sentence.

Tip: process B’ before applying OR.

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BLOCK 2 CHECK

Can you move between all three representations?

QUESTION 1

INPUT VALUES

A = 1 · B = 0

AND = ? OR = ?

QUESTION 2

CIRCUIT → EXPRESSION

A enters AND directly.
B passes through NOT, then AND.

Write Y = _____

Evidence standard

Give the output and one verbal justification—not only a memorised row.

Primitive gates become the building blocks of combinational datapaths.

What happens when a gate’s output is inverted?

AND outputs 0 0 0 1 invert → _____ INVERTED OUTPUTS

OR outputs 0 1 1 1 invert → _____ INVERTED OUTPUTS

Two familiar functions are about to get new names.

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NAND and NOR invert basic gate outputs

NAND



$Y = (A \cdot B)'$

0 only when A = 1 and B = 1

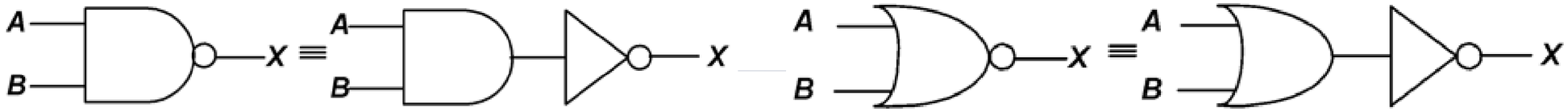
NOR



$Y = (A + B)'$

1 only when A = 0 and B = 0

A	B	NAND	NOR
0	0	1	1
0	1	1	0
1	0	1	0
1	1	0	0



XOR detects difference; XNOR detects equality

XOR $Y = A \oplus B$ 1 when A ≠ B **DIFFERENT**

XNOR $Y = (A \oplus B)'$ 1 when A = B **SAME**

A	B	XOR	XNOR
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

Universal gates rebuild AND and OR algebraically

NAND FAMILY



T = A NAND B
Y = T NAND T = A · B

NOR FAMILY



T = A NOR B
Y = T NOR T = A + B

Diagrams show Stage 1; equations complete Stage 2 with tied T inputs.

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XOR differs from OR in one critical row

FOCUS ROW A = 1 · B = 1

OR	XOR	XNOR
1	0	1
at least one is 1	inputs are same	inputs match

OR means ‘one or both’. XOR means ‘different’ for two inputs.

One XOR/XNOR pair compares two bits

COMPARE TWO BITS

DIFF **= A ⊕ B**

1 when the two bits differ

SAME **= (A ⊕ B)'**

1 when the two bits match

Exactly one output is 1 for every row.

A	B	XOR (DIFFER)	XNOR (SAME)
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

Equality checking is one small ingredient in later ALU and control logic.

PAIR ACTIVITY · 12 MIN

Replace and predict

1. Replace AND + output inversion with one gate. Name it.
2. Replace OR + output inversion with one gate. Name it.
3. Predict NAND, NOR, XOR and XNOR for input pairs 11 and 10.

Justify with output conditions—not memory alone.

BLOCK 3 CHECK

Name the gate from its output condition

A	B	NAND	NOR	XOR	XNOR
0	0	1	1	0	1
0	1	1	0	1	0
1	0	1	0	1	0
1	1	0	0	0	1

0 only for 11

GATE: _____

1 only for 00

GATE: _____

1 when inputs differ

GATE: _____

1 when inputs match

GATE: _____

Forward link: these functions support later ALU, enable and control decisions.

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A truth-table rule applies at every moment

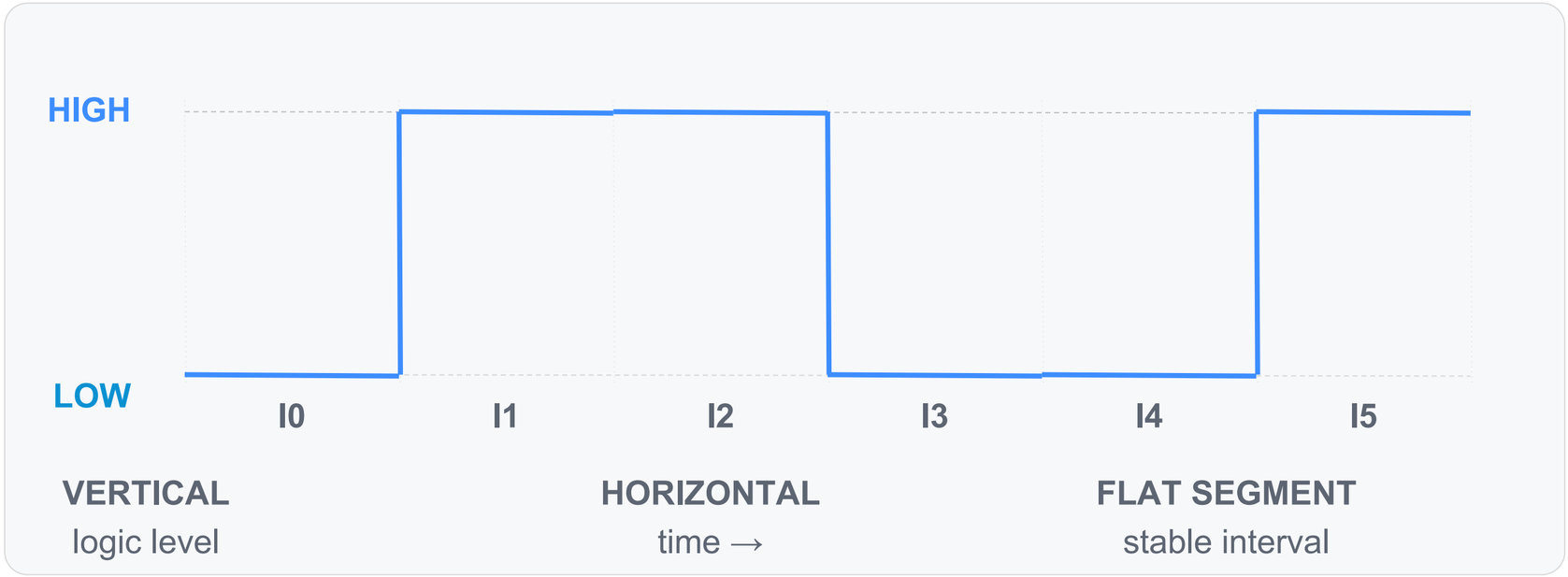
At each stable interval:

1. **READ** — A and B
2. **APPLY** — the gate rule
3. **DETERMINE** — Y

A truth-table row describes the current input condition—not a permanent state.

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A waveform shows logic state against time

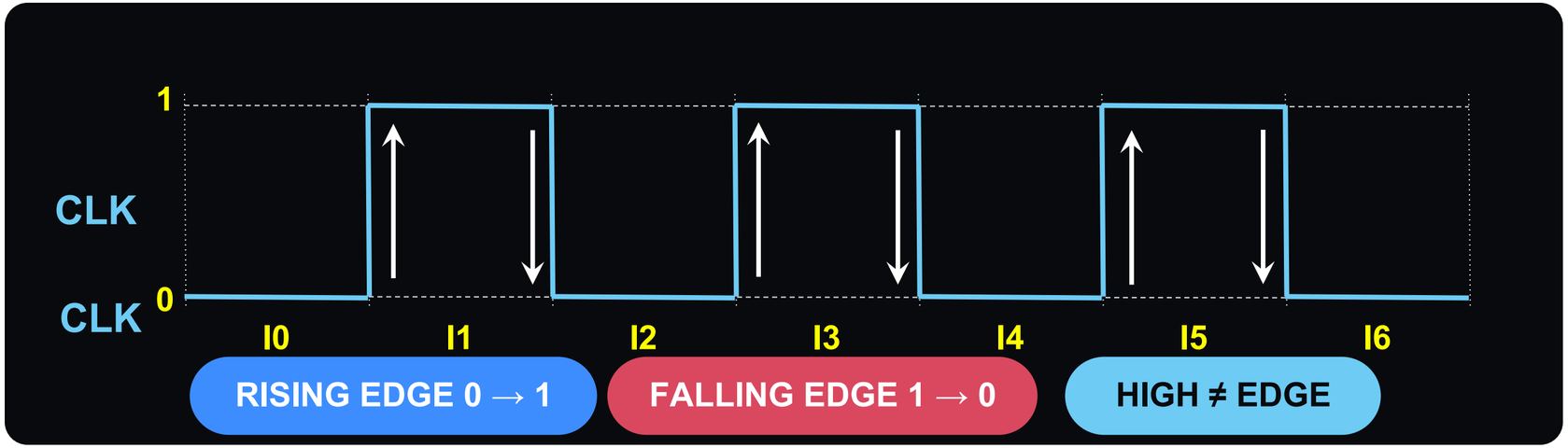


Read intervals—not isolated points.

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Clock edges name the direction of change

Clock behavior	What is happening?
0 → 1	Rising **edge**
1 → 1	Staying **HIGH**; no edge
1 → 0	Falling **edge**
0 → 0	Staying **LOW**; no edge



Propagation awareness

A physical gate responds shortly after an input change. In Week 1 traces, assume ideal zero-delay unless stated.

One pulse: rising edge → HIGH interval → falling edge. Storage comes in Week 9.
HIGH describes a state. An **edge** describes a change

Trace a circuit one stable interval at a time

1 Mark transitions

Every input change creates a boundary.

2 Divide time

Work between boundaries.

3 Read A, B

Example pair: 10.

4 Apply rule

Use the matching truth-table row.

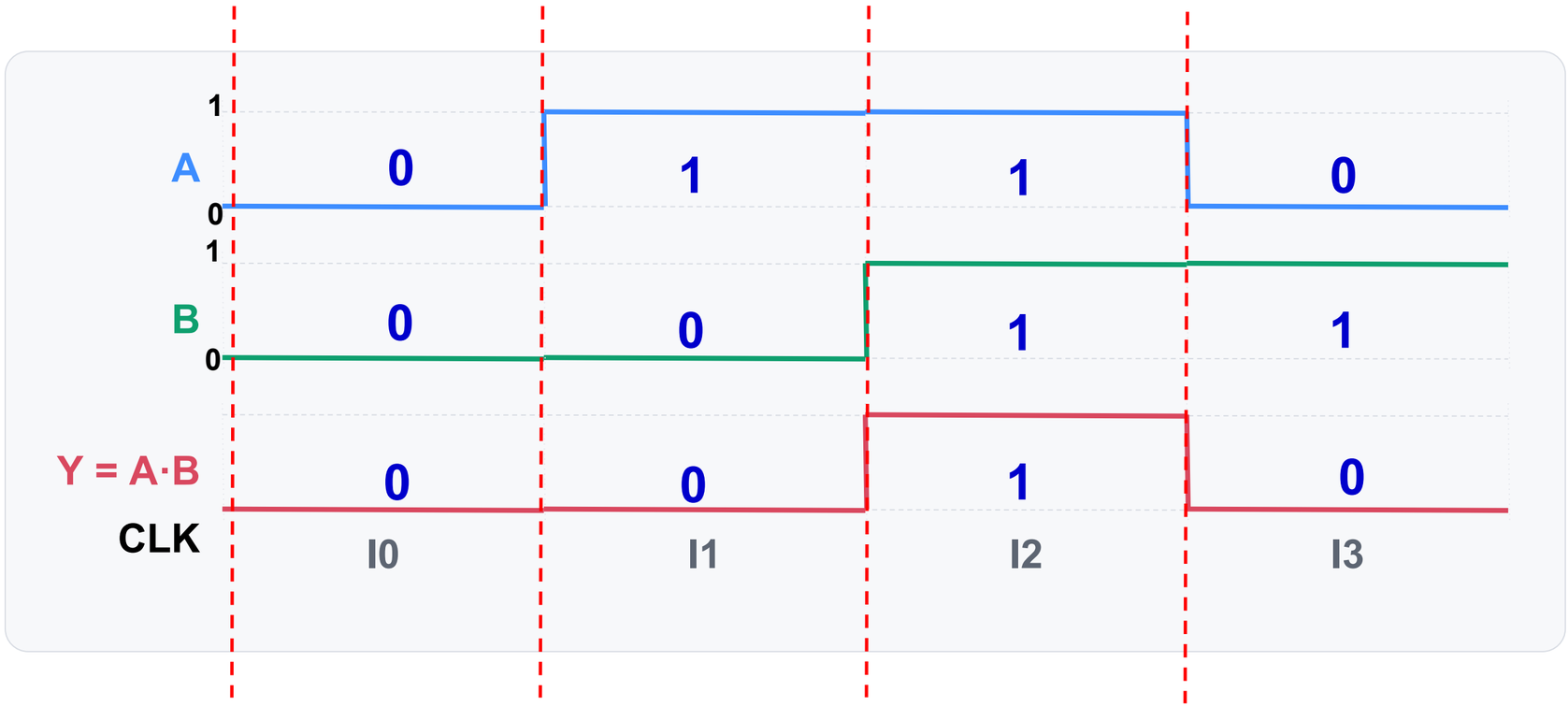
5 Draw Y

Keep Y stable inside the interval.

Intermediate nodes first → final output second.

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Worked trace: AND is HIGH only during 11

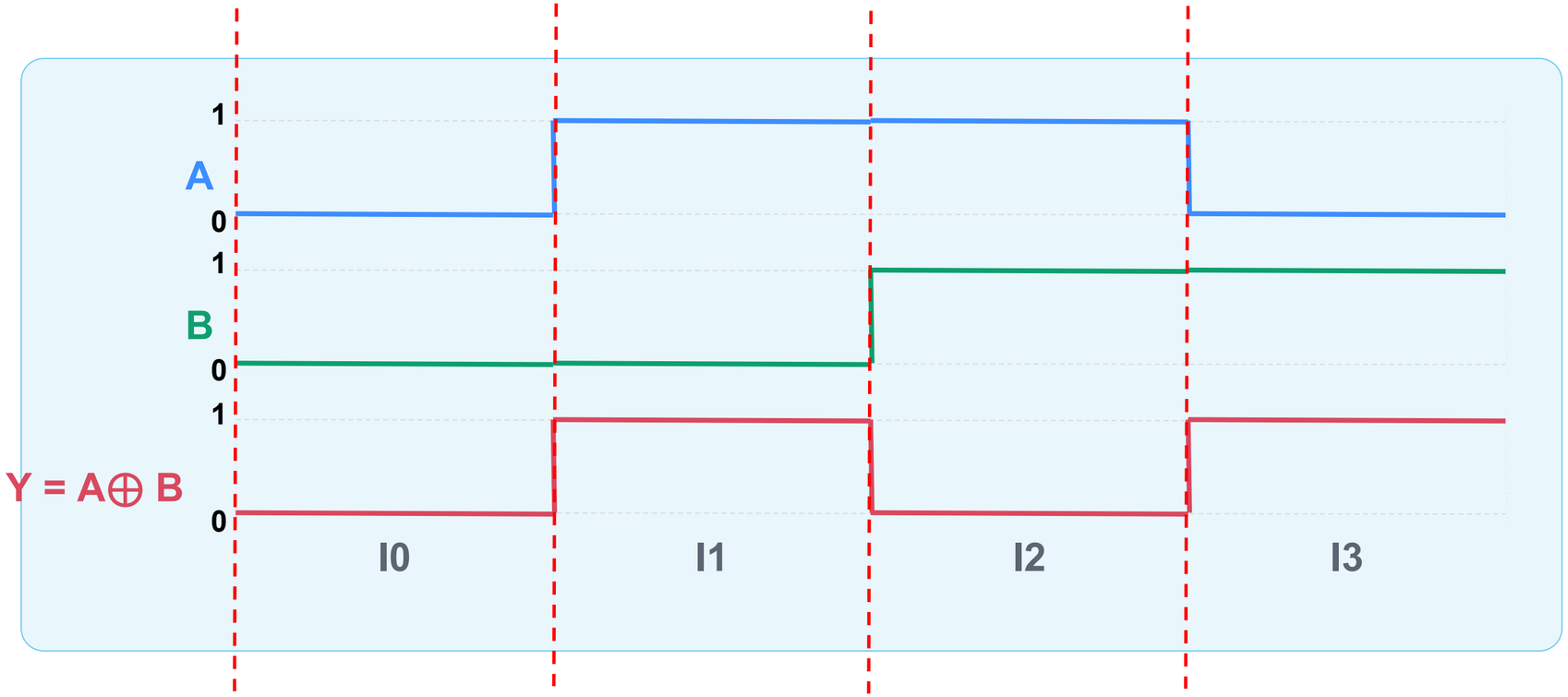


A	B	Y (AND)
0	0	0
0	1	0
1	0	0
1	1	1

Only interval I2 contains A = 1 and B = 1.

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Worked trace: XOR is HIGH when inputs differ



A	B	Y (XOR)
0	0	0
0	1	1
1	0	1
1	1	0

XOR falls during I1 because the inputs match.

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INDIVIDUAL → PAIR → SIMULATION · 12 MIN

Complete the table, then sketch two waveforms

INTERVAL	I0	I1	I2	I3	I4
A	0	0	1	1	0
B	0	1	1	0	0
Y AND	?	?	?	?	?
Y XOR	?	?	?	?	?

1 Complete the YAND and YXOR rows.

2 Sketch both waveforms; verify in the simulator.

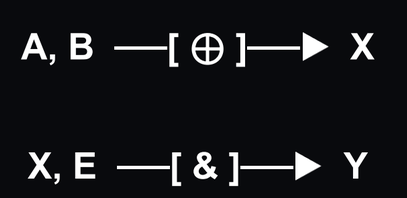
Diagnose interval alignment before gate-rule errors.

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EXIT TICKET · 6 MIN

Week-end checkpoint: trace a small logic system

SMALL CIRCUIT



1. Complete X and Y.
2. Mark Y's 0→1 transition: a rising edge.
3. Explain what caused it.

INTERVAL	I0	I1	I2	I3
A	0	0	1	1
B	0	1	1	0
E	1	0	1	1
X	?	?	?	?
Y	?	?	?	?

Bridge forward **Week 9 revisits these time-varying signals for clocked storage.**

Next in Module 1: Week 2 uses multiple bits to represent numbers, codes and data.

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NUMBER SYSTEMS & CODES

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Week 2 · Data Representation

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WEEK 2 OUTCOMES

By the end of Week 2, you can…

01	**Convert**	accurately among decimal, binary and hexadecimal
02	**Determine**	the range and meaning of an unsigned fixed-width value
03	**Interpret**	BCD, Gray code, ASCII and parity at introductory depth
04	**Relate**	four-bit values to Bridge-4 data, operands and addresses

Mastery: binary, decimal, hexadecimal and fixed width · Survey: selected codes

What does each position contribute?

DECIMAL

347_{10}

Which digit carries hundreds?
Which digit carries ones?

BINARY

101101_2

Which positions are active?
What is the largest weight?

PLACE VALUE

305_{10}

Does the zero disappear—or hold a place?

WIDTH

001011_2

Does a leading zero change the value?
Does it change the width?

Vote first. Then justify using positions—not digit shapes alone.

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A radix defines symbols and place weights

$$\text{value} = \Sigma\,(\text{digit}_i \times \text{base}^i)$$

RADIX 10

Decimal uses base 10

347_{10}

$= 3\times10^2 + 4\times10^1 + 7\times10^0$

$= 300 + 40 + 7$

RADIX 2

Binary uses base 2

101101_2

$= 1\times2^5 + 0\times2^4 + 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0$

The same positional rule works; only the base and allowed digits change.

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Binary places are powers of two

BIT POSITION	5	4	3	2	1	0
	2^5	2^4	2^3	2^2	2^1	2^0
WEIGHT	32	16	8	4	2	1
BIT	1	0	1	1	0	1

Active weights: 32 + 8 + 4 + 1

A 0 holds its position but contributes zero to the sum.

101101_2 means 45_{10}

STEP 1

Expand by position

101101_2

$= 1\times2^5 + 0\times2^4 + 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0$

$= 32 + 0 + 8 + 4 + 0 + 1$

STEP 2

Add the active weights

$32 + 8 + 4 + 1$

$= 45$

$101101_2 = 45_{10}$

Evidence standard: show the expansion, not only the final answer.

Build 45_{10} from the largest powers that fit

1 Start

45 − 32 = 13
Set the 32 bit to 1.

2 Skip 16

16 does not fit 13.
Set the 16 bit to 0.

3 Continue

13 − 8 = 5
5 − 4 = 1

4 Finish

Skip 2; use 1.
Bits: 1 0 1 1 0 1

1	0	1	1	0	1
32	16	8	4	2	1

$45_{10} = 101101_2$

INDIVIDUAL → PAIR · 12 MIN

Convert, compare, and justify

Write the radix on every answer and show one defensible conversion step.

1	**EXPAND**	Convert 110010_2 to decimal using positional weights.
2	**BUILD**	Convert 25_{10} to binary using weights or repeated division.
3	**CHECK**	Convert 37_{10} to binary, then verify by expansion.

A complete answer includes the base label and the conversion path.

BLOCK 1 CHECK

Can you prove what a bit string means?

QUESTION 1

Convert

100111_2 = _____$_{10}$

QUESTION 2

Expand

1101_2 = 8 + _____ + _____

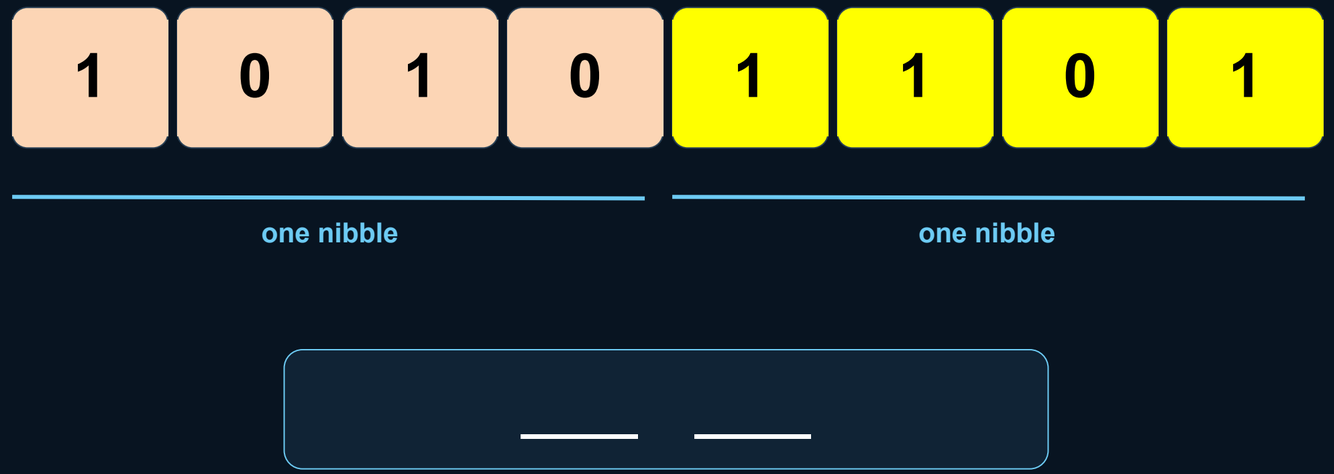
Evidence standard
Give the value, radix labels and one positional-weight line.

Next: four binary bits can be compressed into one hexadecimal digit.

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How would you group 10101101_2?

Start at the right and make groups of four bits.



Predict the two compact labels before seeing the hexadecimal table.

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Hexadecimal compresses four bits into one digit

BINARY

Four binary positions

0000_2 through 1111_2

16 possible patterns

Each pattern fits one hexadecimal digit.

BASE 16

One hex position

0 through 9
A through F

A=10, B=11,C=12,E=13, F=15

$$1010_2 \leftrightarrow A_{16}$$

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Every nibble maps to one hexadecimal digit

BINARY	HEX	DEC
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7

BINARY	HEX	DEC
1000	8	8
1001	9	9
1010	10	A
1011	11	B
1100	12	C
1101	13	D
1110	14	E
1111	15	F

Read the map in both directions: bits ↔ hex ↔ decimal value.

Group from the radix point and preserve width

10101101_2	→	1010 1101	→	AD_{16}
101_2	→	0101	→	5_{16}
00001010_2	→	0000 1010	→	$0A_{16}$
$2F_{16}$	→	0010 1111	→	00101111_2

Leading zero: same value, explicit width.

$1010\ 1101_2$ is AD_{16} and 173_{10}

1	0	1	0	1	1	0	1

$1010 \rightarrow A$ $\qquad$ $1101 \rightarrow D$

$1010\ 1101_2 = AD_{16}$

Convert hex to decimal

$AD_{16} = 10 \times 16^1 + 13 \times 16^0 = 160 + 13 = 173_{10}$

Keep every base label visible while the representation changes.

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$7C_{16}$ is $0111\ 1100_2$ and 124_{10}

HEX → BITS

Translate digits

$7 \rightarrow 0111$

$C \rightarrow 1100$

WIDTH

Preserve groups

$7C_{16}$

$= 0111\ 1100_2$

DECIMAL

Check the value

$7 \times 16 + 12$

$= 112 + 12$

$= 124_{10}$

$7C_{16} = 0111\ 1100_2 = 124_{10}$

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MINI-WHITEBOARD · 12 MIN

Translate across all three bases

Group first, label the base, then check the value.

1 BINARY

$0011\ 1110_2 \rightarrow _____{}_{16} \rightarrow _____{}_{10}$

2 HEX A5

$A5_{16} \rightarrow _____{}_{2} \rightarrow _____{}_{10}$

3 HEX 2D

$2D_{16} \rightarrow _____{}_{2} \rightarrow _____{}_{10}$

Do not merge hex digit values as decimal text.

BLOCK 2 CHECK

Can you keep value and width separate?

QUESTION 1

Translate

$0000\ 1010_2$

$= _____{}_{16}$

$= _____{}_{10}$

QUESTION 2

Explain

Why retain the leading zero nibble?

Evidence standard
Correct grouping, leading-zero handling and explicit base labels.

Meeting 2 asks how many values a declared width can hold.

How many patterns can n bits represent?

1 bit	2 bits	3 bits	4 bits
? patterns	? patterns	? patterns	? patterns

What doubles each time another bit is added?

Count patterns first. Maximum value is a different question.

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n bits create 2^n distinct patterns

1 bit	2 bits	3 bits	4 bits
$2^1 = 2$ patterns	$2^2 = 4$ patterns	$2^3 = 8$ patterns	$2^4 = 16$ patterns
0, 1	00 … 11	000 … 111	0000 … 1111

Each extra bit doubles the number of distinguishable patterns.

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Unsigned range ends at $2^n - 1$

n bits: 0 … $2^n - 1$

3 bits	$2^3 = 8$ patterns	**0 … 7**
4 bits	$2^4 = 16$ patterns	**0 … 15**
8 bits	$2^8 = 256$ patterns	**0 … 255**

Count = 16; maximum = 15. Those statements are both correct.

Width is part of the representation

1

BIT

1 binary position
2 possible states

4

NIBBLE

4 bits
16 possible patterns

8

BYTE

8 bits
256 possible patterns

00001010_2 same value as 1010_2 **but explicitly 8 bits wide**

Leading zeros preserve value and declare storage width.

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Four bits cover every value from 0 to 15

BINARY	HEX	UNSIGNED
1000	8	8
1001	9	9
1010	10	A
1011	11	B
1100	12	C
1101	13	D
1110	14	E
1111	15	F

BINARY	HEX	UNSIGNED
0000	0	8
0001	1	9
0010	2	10
0011	3	11
0100	4	12
0101	5	13
0110	6	14
0111	7	15

16 patterns · minimum 0 · maximum 15

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Four address bits select 16 locations—not address 16

DATA

VALUE FIELD

0000_2 → value 0

…

1111_2 → value 15

One of sixteen immediate values

ADDRESS

ADDRESS FIELD

0000_2 → location 0

…

1111_2 → location 15

One of sixteen locations

MEMORY	ADDRESS
0000	0
0001	1
0010	2
……	…
1111	15

COUNT = 16 · MAXIMUM LABEL = 15

Innovating Sustainable Solutions

PAIR ACTIVITY · 12 MIN

Choose the smallest width that fits

For each value, state the minimum unsigned width and justify it with a range.

1	**SORT**	13, 16, 31 and 32 → minimum widths?
2	**RANGE**	State pattern count and unsigned range for 3, 5 and 8 bits.
3	**FIT CHECK**	Which values do not fit in four bits: 9, 15, 16, 23?

A value fits when it is no greater than $2^n - 1$.

Width means the number of bits. For example, 1101 has a width of **4 bits**.

Innovating Sustainable Solutions

BLOCK 3 CHECK

Can you distinguish count, range and width?

QUESTION 1

Five bits

Patterns: _____

Range: _____

QUESTION 2

Minimum width

Value 31 needs _____ bits.

Value 32 needs _____ bits.

QUESTION 3

Four bits

Does unsigned 16 fit? Explain.

Evidence standard
Justify maximum values and address counts with 2^n and 2^n-1.

Next: the same bits can encode digits, characters or error checks.

ANSWER

1

Decimal value	Binary value	Minimum width	Range at that width
13	1101	**4 bits**	0–15
16	10000	**5 bits**	0–31
31	11111	**5 bits**	0–31
32	100000	**6 bits**	0–63

2

Width	Number of patterns	Unsigned range	Minimum binary	Maximum binary
3 bits	2^3 = **8**	**0–7**	0	111
5 bits	2^5 = **32**	**0–31**	0	11111
8 bits	2^8 = **256**	**0–255**	0	11111111

3

Value	Binary	Fits in four bits?	Reason
9	1001	**Yes**	Within 0–15
15	1111	**Yes**	Exactly the maximum
16	10000	**No**	Needs five bits
23	10111	**No**	Needs five bits

What does 0101 1001 mean?

0	1	0	1	1	0	0	1

UNSIGNED

89

HEX

59_{16}

BCD

decimal 59

ASCII

character ‘Y’

All four can be valid—the contract decides.

Separate the bit pattern from the code chosen to represent information.

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8421 BCD encodes each decimal digit separately

BCD DIGIT

Decimal digit 5

5 → 0101

FOLLOWED BY

BCD DIGIT

Decimal digit 9

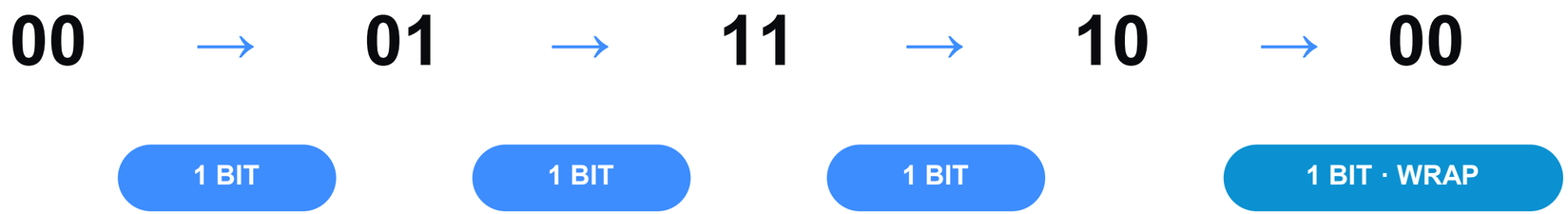
9 → 1001

59_{10} → 0101 1001 BCD

Valid digit codes: 0000–1001 · 1010–1111 are not decimal digits.

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Gray neighbours differ in exactly one bit



Start at 00; every transition—including 10 → 00—changes one bit.

Gray labels organise adjacency; do not evaluate them with binary place weights.

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The same Gray order returns in Week 6 K-maps

CODE ORDER

WEEK 2

Gray sequence

00 · 01 · 11 · 10

Neighbouring labels differ in one bit.



MAP ORDER

WEEK 6

K-map axes

00 · 01 · 11 · 10

Edges wrap to preserve adjacency.

Gray order is not 00 · 01 · 10 · 11

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ASCII names characters; parity checks a bit group

CHARACTER

STANDARD ASCII

‘A’ = 65_{10} = 41_{16}

7-bit code: 1000001_2
8-bit byte: 01000001_2

ERROR CHECK

EVEN PARITY

Data: 0101 1001
Number of 1s: 4

Parity bit p = 0
Append at right: 0101 1001 0
Total remains even.

Parity detects odd-count bit errors; it does not correct them.

One pattern can support several valid meanings

0	1	0	1	1	0	0	1

UNSIGNED

89_{10}

binary place value

HEXADECIMAL

59_{16}

two nibbles

8421 BCD

decimal 59

digit 5 followed by digit 9

ASCII

‘Y’

code point 0x59

Encoding contract first → interpretation second

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INDIVIDUAL → PAIR · 12 MIN

Become a code detective

Name the coding rule before interpreting the bits.

1	**BCD**	Encode decimal 42. Is 1010 a valid single BCD digit?
2	**GRAY**	Complete 00 → 01 → ____ → 10 → 00 and justify adjacency.
3	**PARITY**	Append an even-parity bit to 1010111. State what parity can detect.

ASCII lookup: ‘A’ = 41_{16} = 01000001_2 in an 8-bit byte.

EXIT TICKET · 6 MIN

Week 2 checkpoint: read the pattern and its contract

QUESTION 1

Convert

$0110\ 1001_2$

= _____$_{16}$

= _____$_{10}$

QUESTION 2

Range

Five-bit unsigned range:

_____ to _____

QUESTION 3

Encode

Decimal 37 in 8421 BCD:

_____

Evidence standard
Show base labels, declared width and the coding contract. Gray order: 00 · 01 · 11 · 10.

Week 3 performs arithmetic on these same fixed-width patterns.

BINARY ARITHMETIC

SCSC1013 - COMPUTER SYSTEM & ORGANIZATION

Week 3 · Signed Numbers & Overflow

Dr Zuriahati Mohd Yunos · BSE (Hons.)

WEEK 3 OUTCOMES

By the end of Week 3, you can…

01	**Add**	unsigned binary values and identify carry out
02	**Subtract**	with two’s complement inside a fixed width
03	**Distinguish**	unsigned carry, borrow and signed overflow
04	**Interpret**	four-bit modular arithmetic for Bridge-4 operations

Mastery: fixed-width addition, subtraction and two’s complement

What do these four-bit words mean before we add?

OPERAND A

1011_2

unsigned value = _____

maximum four-bit value = _____

OPERAND B

0110_2

unsigned value = _____

width = _____ bits

Declare width and interpretation before calculating.

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One-bit addition can produce a sum and a carry

A	B	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$1 + 1 = 10_2 \rightarrow$ write 0, carry 1

The carry enters the next more-significant column.

A carry moves left until a column can absorb it

carry		1	1	1
		0	1	1	1
+		0	0	0	1
		1	0	0	0

1 Rightmost

1+1=0; carry 1.

2 Middle

1+carry=0; carry continues.

3 Left

0+carry=1; chain stops.

Work column-by-column from right to left.

Innovating Sustainable Solutions

1011 + 0110 gives stored 0001 with carry 1

FOUR-BIT OPERANDS

Full addition

carry chain

```
      1 1 1
    1 0 1 1
+   0 1 1 0
-----------
  1 0 0 0 1
```

RESULT

Interpret the fixed-width result

Full unsigned sum: 17

Stored low four bits: 0001
Carry-out: C = 1

11 + 6 = 17

register ← 0001 · C ← 1

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A four-bit register stores low bits; carry is separate

stored result = (A + B) mod 16

5 + 3 = 8

No carry out

0101 + 0011 = 1000

Stored: 1000
C = 0

14 + 3 = 17

Carry out

1110 + 0011 = 1 0001

Stored: 0001
C = 1

C records unsigned overflow beyond the declared width.

PAIR ACTIVITY · 12 MIN

Track every carry before recording the result

For each sum, show the carry chain, stored four-bit result and C.

1 **CASE A**

0101 + 0011 → result _____ · C = ___

2 **CASE B**

1110 + 0011 → result _____ · C = ___

3 **CASE C**

0111 + 0001 → result _____ · C = ___

Do not diagnose signed overflow yet; first record the fixed-width facts.

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BLOCK 1 CHECK

Can you separate the sum from carry out?

QUESTION 1

Calculate

1101 + 0101
= _____
C = ___

QUESTION 2

Explain

Where is the fifth bit recorded in a four-bit machine?

Evidence standard
Correct stored sum, complete carry chain and fixed-width interpretation.

Next: subtraction becomes addition of a two’s complement.

How can subtraction become addition?

FAMILIAR

BORROWING

0101 − 0011

Work directly across columns.

OR

MACHINE PATH

COMPLEMENT

0101 + two’s complement of 0011

Use the same addition hardware.

A − B = A + (two’s complement of B)

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Both subtraction paths give the same difference

A − B

DIRECT SUBTRACTION

Compare each bit column.
Borrow when the top bit is too small.

Human-friendly procedure.

A + (−B)

COMPLEMENT ADDITION

Invert B at the declared width.
Add 1.
Add the result to A.

Adder-friendly procedure.

Same low four-bit result · different path

Subtract by forming B’s two’s complement

1 Declare width

Use exactly four bits.

2 Invert B

Flip every bit: 0↔1.

3 Add one

This forms −B mod 16.

4 Add to A

Keep low four bits; record C.

$$-B \text{ in } n \text{ bits} = (\sim B + 1) \bmod 2^n$$

Invert-and-add-one only makes sense at a declared width.

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0101 − 0011 gives 0010 with no borrow

TWO'S COMPLEMENT

Form −0011

0011
↓ invert
1100
↓ add 1
1101

A + (−B)

Add to A

0101
+ 1101
———————
1 0010

Stored: 0010 · C=1

5 − 3 = 2 · C=1 means no unsigned borrow

Innovating Sustainable Solutions

0011 − 0101 wraps to 1110 with borrow

TWO'S COMPLEMENT

Form −0101

0101
↓ invert
1010
↓ add 1
1011

A + (−B)

Add to A

0011
+ 1011
———————
0 1110

Stored: 1110 · C=0

Unsigned: wrap to 14 · Signed: −2 · C=0 means borrow

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Bridge-4 uses C=1 for no unsigned borrow

UNSIGNED RELATION	C	BORROW	INTERPRETATION
A >= B	1	0	No unsigned borrow
A < B	0	1	Borrow, result wraps mod 16

Locked course convention

borrow = 1 − C · other processors may name or invert subtraction flags differently

Always state the convention before interpreting subtraction C.

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INDIVIDUAL → PAIR · 12 MIN

Find the error before trusting the result

Show the four-bit complement of B, the stored result and Bridge-4 C.

1 **CASE A**

1001 − 0011 → result _____ · C = ___

2 **CASE B**

0010 − 0111 → result _____ · C = ___

3 **CASE C**

0000 − 0001 → result _____ · C = ___

Audit width, inversion, add-one and carry interpretation in that order.

BLOCK 2 CHECK

Can you report result and borrow convention?

QUESTION 1

No borrow?

0110 − 0010
= _____
C = ___

QUESTION 2

Borrow?

0100 − 0110
= _____
C = ___

Evidence standard
Correct two’s-complement formation, stored result and explicit C=1 no-borrow rule.

Meeting 2 reinterprets the same patterns as signed two’s-complement values.

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Is 1110 fourteen—or minus two?

1	1	1	0

UNSIGNED	TWO'S COMPLEMENT	HEXADECIMAL
14	−2	E

The bits are unchanged; the interpretation is declared.

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Two’s complement is the signed mastery route

4-BIT INTERPRETATION	RANGE	ZEROS	ROLE HERE
Unsigned	0 …... 15	one	Week 2 mastery
Sign-and-magnitude	-7 ….. +7	two	Comparison only
One’s complement	-7 …. +7	two	Comparison only
Two’s complement	-8 …. +7	one	Week 3 mastery

One zero · simple addition path · asymmetric range

Sign-and-magnitude and one’s complement remain awareness-only.

Additional note

Type	Meaning	Example range with 4 bits
Unsigned integer	Zero and positive whole numbers	**0 to 15**
Signed integer	Negative numbers, zero, and positive whole numbers	**–8 to +7**, using two's complement

Integer type	Minimum value	Maximum value
Unsigned	0	$2^n - 1$
Signed (two's complement)	$-2^n - 1$	$2^{n-1} - 1$

Four-bit two’s complement spans −8 to +7

NON-NEGATIVE

MSB = 0

Unsigned code u = 0…7
Signed value = u

0000 → 0
0111 → +7

NEGATIVE

MSB = 1

Unsigned code u = 8…15
Signed value = u − 16

1000 → −8
1111 → −1

$$-2^3 \ldots 2^3-1 \;=\; -8 \ldots +7$$

The negative side has one extra value: −8 has no +8 partner in four bits.

Encode −3 by complementing +3 at four bits

1 Write +3

0011

2 Invert

1100

3 Add one

1100 + 0001

4 Result

−3 = 1101

1	1	0	1
sign			

$-3_{10} = 1101_2$ in four-bit two's complement

Innovating Sustainable Solutions

Decode 1010 as −6

u − 16

METHOD 1 · CODE VALUE

1010_2 = 10 unsigned

MSB = 1, so
10 − 16 = −6

COMPLEMENT

METHOD 2 · MAGNITUDE

1010
↓ invert
0101
↓ add 1
0110 = 6
Therefore value = −6

$1010_2 = -6_{10}$

Innovating Sustainable Solutions

Sign extension preserves the signed value

4 BITS	VALUE	8 BITS	VALUE
0101	+5	0000 0101	+5
1101	-3	1111 1101	-3
1101	-3	0000 1101	+13

WRONG

Unsigned → extend with 0 · Signed → replicate the sign bit

1101 → 11111101 preserves −3; zero extension changes the signed meaning.

INDIVIDUAL → PAIR · 12 MIN

Encode, decode, and order signed values

Use four-bit two’s complement and show one conversion step.

1. **ENCODE** — −1, −5 and −8 → four-bit patterns
2. **DECODE** — 1100 and 1001 → signed decimal values
3. **ORDER** — 1000, 1101, 0010, 0111 from smallest to largest

Check the allowed range −8…+7 before encoding.

BLOCK 3 CHECK

Can you preserve signed meaning across widths?

QUESTION 1

Interpret

1011_2 = _____$_{10}$
(four-bit signed)

QUESTION 2

Extend

1011 → _____ _____
(eight-bit signed)

QUESTION 3

Range

Four-bit two's-complement:
_____ to _____

Evidence standard
Correct signed range, sign interpretation and value-preserving extension.

Block 4 asks whether a correct bit pattern represents an out-of-range signed result.

Does carry prove signed overflow?

CARRY

CASE A

1110 + 0011
= 1 0001

C = 1
Signed: −2 + 3 = 1

NO CARRY

CASE B

0101 + 0100
= 0 1001

C = 0
Signed: 5 + 4 = 9?

Which case exceeds the signed range −8…+7?

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Carry and overflow answer different questions

UNSIGNED

CARRY OUT · C

Unsigned question

Did A + B exceed 15?

C = 1 when full unsigned sum ≥ 16.

CONCEPTUAL

SIGNED OVERFLOW CONDITION

Signed question

Did the true signed result leave −8…+7?

Same-sign inputs + opposite-sign result.

Either, both, or neither can occur

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A sign flip reveals same-sign addition overflow

+ + → −	**OVERFLOW**	Two positives cannot produce a valid negative sum.
− − → +	**OVERFLOW**	Two negatives cannot produce a valid non-negative sum.
+ − → ±	**NO ADDITION OVERFLOW**	Opposite signs move the sum toward the representable centre.

signed overflow (addition) = same input signs AND opposite result sign

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Carry and signed overflow are independent

OPERANDS	STORED	C	SIGNED OVERFLOW?	SIGNED READING
0010 + 0011	0101	0	0	2 + 3 = 5
0111 + 0001	1000	0	1	7 + 1 is out of range
1111 + 0001	0000	1	0	-1 + 1 = 0
1000 + 1000	0000	1	1	-8 +8 is out of range

C / overflow combinations: 00 · 01 · 10 · 11

Report C and signed overflow independently.

Bridge-4 stores Z and C—but no N or V

BRIDGE-4

IMPLEMENTED FLAGS

When an instruction updates Z:
Z = 1 iff stored ACC result is 0000

ADDI: C = carry-out
SUBI: C = 1 means no borrow

BOUNDARY

NOT IMPLEMENTED

No N flag
No V flag
No JC instruction

JZ is the only conditional branch in scope.

Recognise signed overflow conceptually; do not invent a Bridge-4 V flag.

Diagnose fixed-width arithmetic in four steps

1 Declare

Width and signed/unsigned interpretation.

2 Calculate

Keep the full sum or complement-add path.

3 Record

Store low four bits and C separately.

4 Interpret

Check unsigned range, borrow rule and signed overflow.

Subtraction reminder

Bridge-4: C=1 means no unsigned borrow; signed subtraction overflow is a separate concept.

Compute first. Interpret second. Report each status independently.

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INDIVIDUAL → PAIR · 12 MIN

Diagnose result, C, and signed overflow

Cases A to C: treat C and signed overflow as independent reports. Case D: report result and C only.

1	CASE A	1110 + 0011 → stored ____ · C __ · overflow? ____
2	CASE B	0101 + 0100 → stored ____ · C __ · overflow? ____
3	CASE C	1001 + 1010 → stored ____ · C __ · overflow? ____
4	CASE D	1000 − 0001 → stored ____ · C __

For subtraction, C=1 means no unsigned borrow under the locked convention. Signed overflow is judged for addition only in Week 3.

EXIT TICKET · 6 MIN

Week 3 checkpoint: calculate, then interpret

QUESTION 1

ADDITION

0110 + 0011

Stored = _____
C = ___
Signed overflow? _____

QUESTION 2

SUBTRACTION

0011 − 0101

Stored = _____
C = ___
Borrow? _____

QUESTION 3

EXPLAIN IN ONE SENTENCE

Why can carry and signed overflow give different answers for the same addition?

Next in Module 2: Boolean algebra turns bit-level rules into simplified logic.